

A “torn bag mechanism” of small extracellular vesicle release via limiting membrane rupture of *en bloc* released amphisomes (amphictosomes)

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eLife Assessment

In this study, the authors present **compelling** data illustrating a potential mechanism for a hitherto not described form of extracellular vesicle biogenesis. Their model suggests that small extracellular vesicles are secreted from cells within larger vesicles, termed amphictosomes, which subsequently rupture to release their smaller vesicle contents. This discovery represents an **important** advancement in the field.

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Abstract

Recent studies showed an unexpected complexity of extracellular vesicle (EV) biogenesis pathways. We previously found evidence that human colorectal cancer cells *in vivo* release large multivesicular body-like structures *en bloc*. Here, we tested whether this large extracellular vesicle type is unique to colorectal cancer cells. We found that all cell types we studied (including different cell lines and cells in their original tissue environment) released multivesicular large EVs (MV-IEVs). We also demonstrated that upon spontaneous rupture of the limiting membrane of the MV-IEVs, their intraluminal vesicles (ILVs) escaped to the extracellular environment by a “torn bag mechanism”. We proved that the MV-IEVs were released by ectocytosis of amphisomes (hence, we termed them amphictosomes). Both ILVs of amphictosomes and small EVs separated from conditioned media were either exclusively CD63 or LC3B positive. According to our model, upon fusion of multivesicular bodies with

autophagosomes, fragments of the autophagosomal inner membrane curl up to form LC3B positive ILVs of amphisomes, while CD63 positive small EVs are of multivesicular body origin. Our data suggest a novel common release mechanism for small EVs, distinct from the exocytosis of multivesicular bodies or amphisomes, as well as the small ectosome release pathway.

Introduction

Extracellular vesicles (EVs) are phospholipid bilayer enclosed structures^{1,2,3,4}, which have important roles in cellular homeostasis and intercellular communication. Exosomes have been defined as small (~ 50-200 nm) EVs (sEVs) of endosomal origin^{1,3,4,5}. Although autophagy is a major cellular homeostatic mechanism, and is implicated in a broad spectrum of human diseases, the intersection of autophagy and exosome secretion remains poorly understood. Recently, regulatory interactions have been shown between autophagy-related molecules and EV biogenesis^{5,6}. Furthermore, the LC3-conjugation machinery was demonstrated to specify the cargo packaged into EVs⁷. Importantly, both others and we reported the secretion of LC3-carrying exosomes^{7,8}. Particularly relevant to the findings presented here, is the implication of amphisomes (hybrid organelles formed by the fusion of late endosomes/multivesicular bodies (MVBs) with autophagosomes^{9,10}) in EV biogenesis. It was suggested that fusion of the limiting membrane of amphisomes with the plasma membrane of cells results in a subsequent release of exosomes by exocytosis^{1,3,11}. The current study was prompted by our recent data showing the *in vivo en bloc* release of large, MVB-like sEV clusters by human colorectal cancer cells¹². Here we investigated if this was a colorectal cancer cell-specific phenomenon. Unexpectedly, we found that it was a general mechanism of sEV release that we designated as “torn bag mechanism”.

Results and Discussion

In this study, we analyzed *in situ* fixed, cultured cells with the released EVs preserved in their original microenvironment on a surface coated by gelatin and fibronectin. We detected large multivesicular EVs (MV-IEVs) in sections of different immersion fixed organs. We tested tumorous HT29, HepG2, and non-tumorous HEK293, HEK293T-PalmGFP, HL1 cell lines, as well as primary suspension type bone marrow derived mast cells (BMMCs). In addition, we studied ultrathin sections of mouse kidney and liver.

By the analysis of transmission electron micrographs of all tested cell types, we identified budding (Fig.1A-G) and secretion (Fig.1H-N) of MV-IEVs carrying intraluminal vesicles (ILVs). Importantly, in all cases we found evidence for the extracellular rupture of the limiting membrane of MV-IEVs and the release of ILVs (Fig.1O-U). For this novel type of sEV release, we suggest the designation “torn bag mechanism”, which is distinct from the exocytosis of MVBs and amphisomes^{1,3,4,11} and from the release of plasma membrane derived small EVs (sEVs) by ectocytosis¹³.

Most relevant to the *in vivo* conditions, we also observed the same phenomenon within the ultrathin sections of both murine kidney (Fig.1V) and liver (Fig.1W,X). In these cases, both the intact MV-IEVs (Fig.1V-X) and the “torn bag release” of sEVs (Fig.1V) were detected. Fig.1X shows that a circulating leukocyte releases MV-IEVs by ectocytosis. In Fig.1_S1, MV-IEVs, IEVs and sEVs were captured simultaneously both in kidney (Fig.1_S1C,D) and liver (Fig.1_S1F,H,I). In the mouse liver section (Fig.1_S1G), MV-IEV secretion both by endothelial and subendothelial cells can be detected.

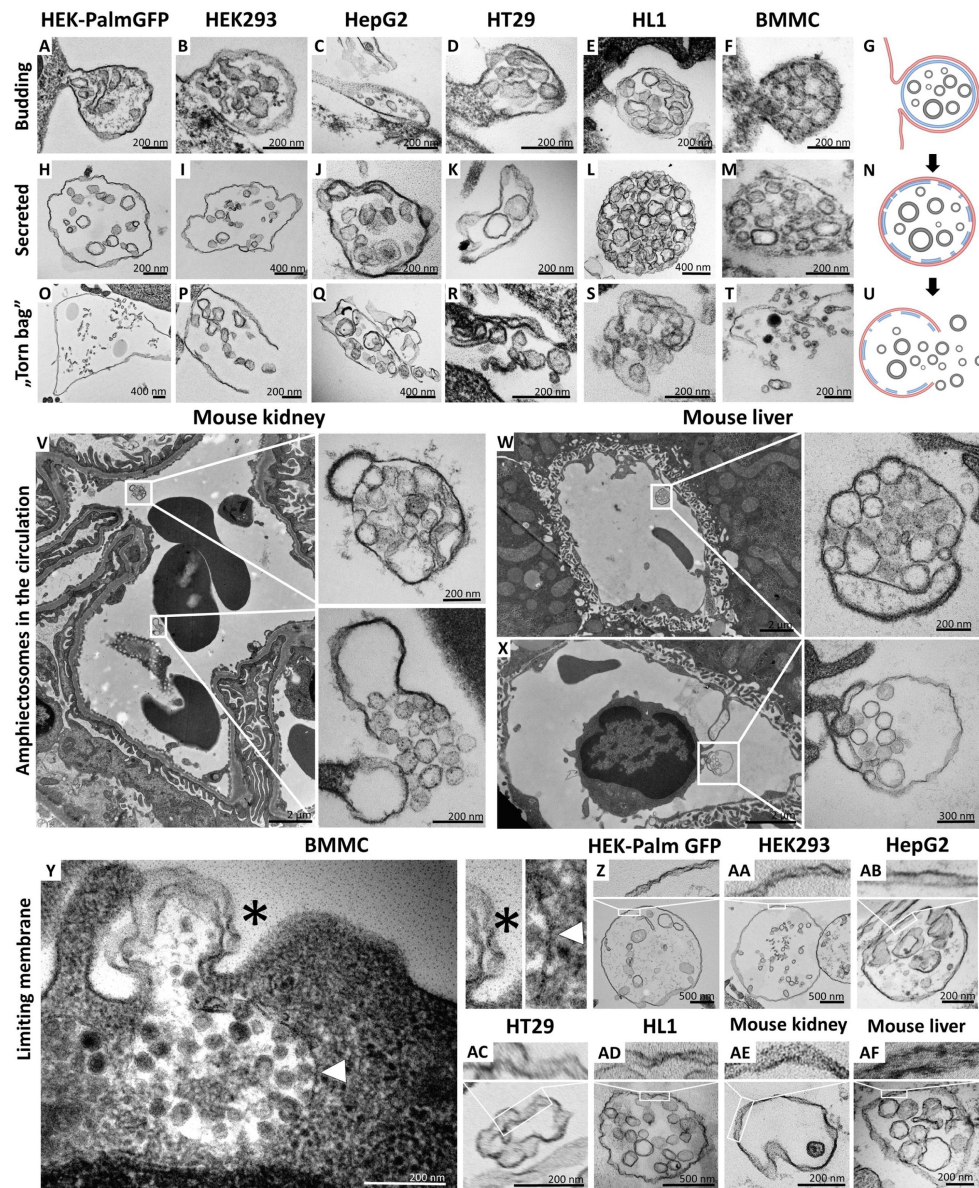


Figure 1.

Transmission electron microscopic detection of the release and extracellular fate of large, multivesicular EVs secreted by different cell lines and cells in mouse organs

Major steps of the release of large, multivesicular EVs (MV-IEVs) were detected in the case of all tested cell lines including the immortal, non-tumorous HEK293T-PalmGFP (A,H,O), HEK293 (B,I,P), the tumorous cell lines HepG2 (C,J,Q) and HT29 (D,K,R), the beating cardiomyocyte cell line HL1 (E,L,S) and the primary suspension of bone marrow derived mast cells (BMMCs) (F,M,T). The different phases of EV secretion were also captured in the circulation of mouse kidney (V) and liver (W,X). According to the electron micrographs, we found evidence for the budding (A-G,X) and secretion (H-N,V,W) of the MV-IEVs. We also detected the extracellular rupture of the limiting membrane of the released MV-IEVs with the escape of the intraluminal vesicles (ILVs) by a "torn bag mechanism" (O-U,V). Although it is not always clear whether the secreted MV-IEVs have a single or double limiting membrane, several micrographs suggest the presence of the double membrane (Y-AF) in the secreted MV-IEVs. In the case of BMMCs (Y), the release phase of a multivesicular structure is captured. The bottom portion of this structure embedded in the cytoplasm is surrounded by a single membrane (white arrow head) while the upper (budding) portion is covered by double membrane (asterisk). In the schematic figures (G, N, U) the limiting membrane of MV-IEV presumably with plasma-membrane origin was indicated by red, the original limiting membrane of intracellular amphisomes, which may fragmented during the release process was indicated by blue while the ILVs of the MV-IEV were shown by gray color.

Based on the transmission electron microscopy (TEM) analysis of ultrathin sections, it was not always obvious whether the secreted MV-IEVs had a single or double membrane. However, several micrographs suggested an at least partially intact double membrane (**Fig.1Y-AF**), of MV-IEVs. In the case of BMBCs (**Fig.1Y**), the release phase of a multivesicular structure is captured. The bottom portion of this structure, embedded in the cytoplasm, is surrounded by a single membrane while the upper (budding) portion is covered by double membrane. We hypothesize that disruption of the original amphisome membrane mainly occurs after separation of the MV-IEV from the cell to avoid the release of ILVs inside the cell.

Next, we decided to further investigate the subcellular origin of the ILVs within the secreted MV-IEVs. First, we analyzed the microenvironment of *in situ* fixed HEK293T-PalmGFP cells by confocal microscopy. The PalmGFP signal of HEK293T-PalmGFP cells principally associates with the plasma membrane¹⁴ (**Fig.2_S1A-C**), therefore the green fluorescence helped us to identify the plasma membrane-derived limiting membrane of MV-IEVs. In agreement with our previous findings on HT29 colorectal cancer cells, within the MV-IEVs, we found CD63/ALIX (**Fig.2A,G**), CD81/ALIX (**Fig.2B,H**), CD63/TSG101 (**Fig.2C,I**) and CD81/TSG101 (**Fig.2D,J**) double positive ILVs or ILV clusters.

We also studied the possible autophagy related aspects of the secreted MV-IEVs. ILVs were tested for the autophagy marker LC3B in parallel with CD63 and CD81. Although LC3B, CD63 and CD81 were all present in association with the ILVs (**Fig.2E,F**), the LC3B and CD63 (**Fig.2K**) and the LC3B and CD81 (**Fig.2L**) signals did not overlap. **Fig.2M** shows that while the known sEV markers (CD63, CD81, TSG101 and ALIX) strongly co-localized with each other, LC3B positivity hardly showed colocalization with CD63 or CD81. Immunocytochemistry analysis of HT29, HepG2 and the cardiomyoblast H9c2 cells further validated the findings obtained with the HEK293T-PalmGFP cells (**Fig.2_S2** and **Fig.2_S3A,B**). The ILVs of HEK293T-PalmGFP and HepG2 cell lines were also Rab7 positive (**Fig.2_S3C,D**), suggesting a late endosomal origin. Western blotting of the applied antibodies is summarized in **Fig.2_S4**.

The sEV markers were also tested by TEM using negative-positive contrasting technique¹⁵ on sEVs separated from serum-free conditioned medium of HEK293T-PalmGFP cells. **Fig.2P** confirms the typical sEV morphology. With TEM double immunogold labelling, using anti-LC3B and anti-CD63 antibodies simultaneously, we found distinct LC3B positive (**Fig.2Q**, **Fig.2_S6O**) and CD63 positive (**Fig.2R**, **Fig.2_S6O**) sEVs. Based on the analysis of TEM images, the diameters of unlabeled, and LC3B positive and negative sEVs were determined (**Fig.2S**). The LC3B positive sEVs had a significantly larger diameter as compared to the LC3B negative ones.

To conclude our marker studies, we detected the presence of CD63, CD81, TSG101, ALIX positive, most probably MVB-derived ILVs. In addition, the autophagosome marker carrying, LC3B positive ILVs were also found within the same single, plasma membrane limited extracellular MV-IEV, which identified these MV-IEVs as *en bloc* released amphisomes¹⁶ that we refer to “amphictosomes”.

The “torn bag mechanism” was also monitored by live cell SIM² super-resolution microscopy analysis of HEK293T-PalmGFP-LC3RFP cells (**Fig.2N,O**). The release of the LC3 positive red fluorescent signal was detected within a relatively short period of time (the first LC3 positive ILVs left the amphictosome within 40s, the whole “torn bag” sEV release process was completed within 260 sec) (**Fig.2O**). We could rule out the possibility that rupture of the limiting membrane detected by TEM (**Fig.1O-T,V**) was a fixation artefact by showing the spontaneous release of LC3 positive sEVs from amphictosomes with live-cell imaging. Characterization of the in-house developed HEK293T-PalmGFP-LC3RFP cell line is shown in **Fig.2_S5**.

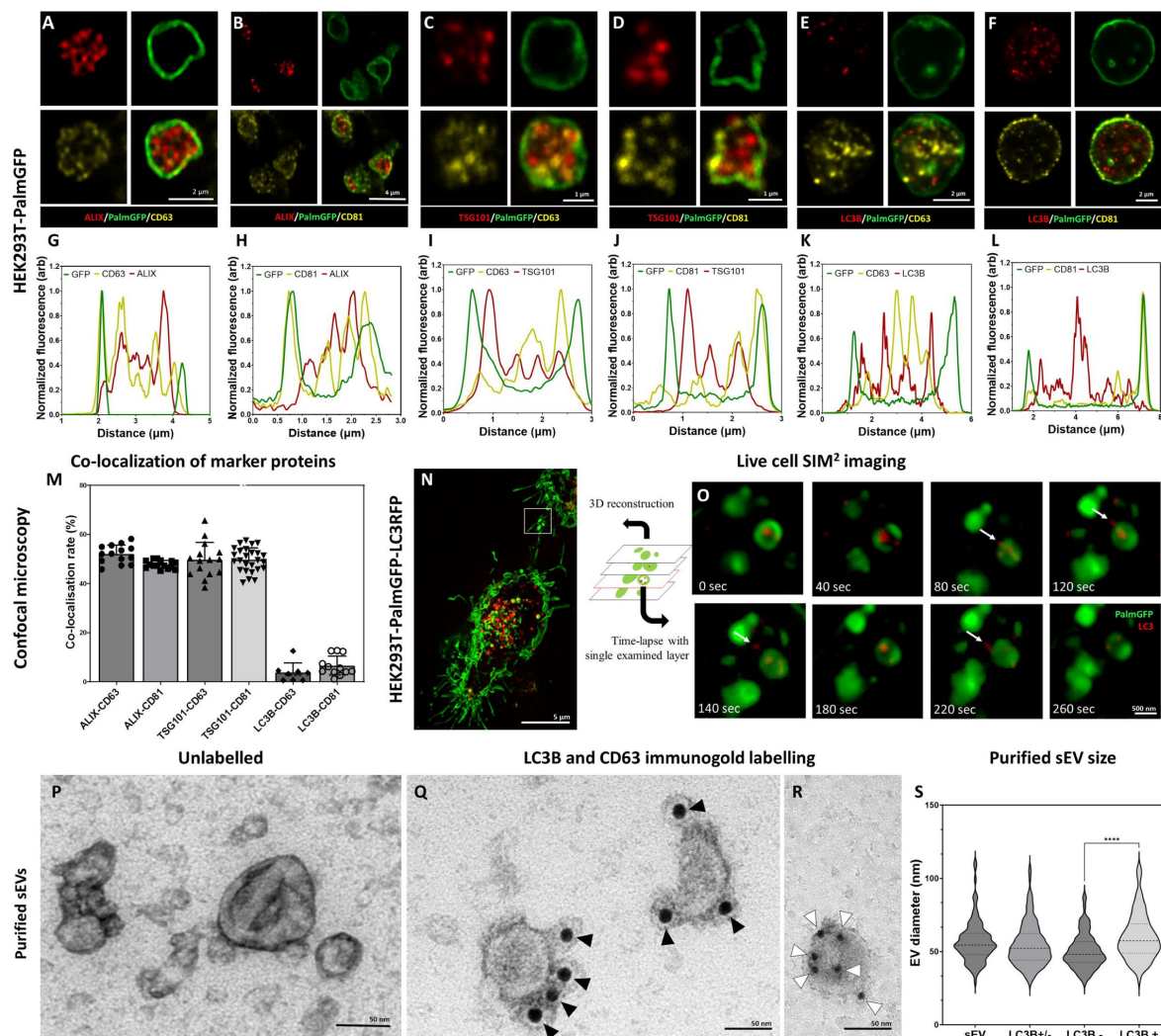


Figure 2.

Detection of conventional sEV markers and the LC3 protein in HEK293T-PalmGFP cell-derived EVs

Widely used sEV markers (CD63, CD81, ALIX and TSG101) and LC3B were tested in MV-IEVs found in the microenvironment of the releasing cells by confocal microscopy after *in situ* fixation (A-F). Normalized fluorescence intensities were calculated in order to determine the relative localization of the limiting membrane (PalmGFP), the conventional sEV markers and the LC3B signal (G-L). Fluorescence intensity peaks of sEV markers were largely overlapping with each other, while the LC3B signal and the sEV markers showed separation. Co-localization rates were also calculated (M). The sEV markers co-localized with one another as no significant difference was found among them. In contrast, low co-localization rates were detected between the “classical” sEV markers and LC3B (one-way ANOVA, $p < 0.0001$, $n = 8-26$ confocal images). Real time release of LC3 positive sEVs by the “torn bag mechanism” was studied in the case of HEK293T-PalmGFP-LC3RFP cells by Elyra7 SIM² super-resolution live cell imaging (N,O). Images were recorded continuously, and selected serial time points are shown. LC3 positive, red fluorescent small particles were released within a 5 min timeframe (O) and are indicated by white arrows. Presence of CD63 and LC3B were detected in the case of a sEV fraction separated from serum-free condition medium using immunogold transmission electronmicroscopy (TEM). HEK293T-PalmGFP derived sEV fraction is shown by negative-positive contrast without immune labelling (P). In double-labelled immunogold TEM images (Q,R), distinct LC3B positive (Q) and CD63 positive (R) sEVs were found. However, CD63-LC3B double positive EVs were not detected. Black arrowheads indicate 10 nm gold particles identifying LC3B, while white arrowheads show 5 nm gold particles corresponding to the presence of CD63. Quantitative analysis of TEM images was performed (S), and the diameters of different EV populations were determined. The LC3B negative population was significantly smaller than the LC3B positive one ($p < 0.0001$, t-test; $n = 79-100$). No difference was detected when the immunogold labelled sEV fraction (either LC3B positive or negative, LC3B+/-) and the unlabeled sEV fraction (sEV) were compared ($p < 0.05$, t-test, $n = 112-179$).

In the following step, we addressed the question whether LC3, associated with the ILVs of MV-IEVs, indeed reflected autophagy origin. We tested MVBs (Fig.2_S6A,F,K), autophagosomes (Fig.2_S6B,G,L), amphisomes (Fig.2_S6C,H,M), amphiectosomes (Fig.2_S6D,I,N) and isolated sEV fractions of the same cells (Fig.2_S6E,J,O). Using immune electron microscopy, as expected, we found CD63 single positivity in MVBs (Fig.2_S6K). In autophagosomes, the limiting phagophore membrane was LC3B positive, and CD63 positivity was also present (Fig.2_S6L). The limiting membrane of amphisomes was LC3B negative, and the internal membranous structures were either LC3B or CD63 positive (Fig.2_S6M). The same immunoreactivity was also observed in the ILVs of the released amphiectosomes (Fig.2_S6N). Importantly, sEVs separated from serum-free conditioned medium of HEK293T-PalmGFP cells were either LC3B or CD63 positive (Fig.2_S6O). Thus, we confirmed our confocal microscopy results at the ultrastructural level. Using immunogold TEM, we provided further evidence for the budding/ectocytosis mechanism of amphiectosome release (Fig.2_S6P). The diameters of ILVs within MVBs, amphisomes and amphiectosomes were compared (Fig.2_S6Q), and the differences were likely due to the different membrane composition, pH and osmotic conditions within these structures. In agreement with our observations with separated sEVs, LC3B positive ILVs had a significantly larger diameter than the LC3B negative ones (Fig.2_S6R) possibly indicating difference in membrane composition and their different intracellular origin. Based on all the above findings, we propose the following model (**Fig.3A**): autophagosomes and MVBs fuse to form amphisomes, and the inner, LC3 positive membrane of autophagosomes undergoes fragmentation¹⁶. Membrane fragments curl up and form LC3 positive ILVs. Therefore, amphisomes contain both MVB-derived CD63 positive/LC3 negative and autophagosome-derived, CD63 negative/LC3 positive ILVs. The amphisome is next released from the cell by ectocytosis. Finally, the plasma membrane-derived outer membrane ruptures enabling the ILVs escape to the extracellular space by a “torn bag mechanism”. By using stimulated emission depletion (STED) microscopy, we documented the intracellular phases of our proposed model: MVB (**Fig.3.B**), autophagosome (**Fig.3.D**), the fusion of MVB and autophagosome (**Fig.3.C**), fragmentation of the LC3 positive membrane and ILV formation from the membrane fragments (**Fig.3.F**) and mature amphisome (**Fig.3E**). The plasma membrane origin of the external membrane of amphiectosome was further supported by wheat germ agglutinin-based live cell labelling (**Fig.3G**).

To investigate the process of amphiectosome release, we exposed the MV-IEV releasing cells to different *in vitro* treatments (**Fig.3H**). The release of MV-IEVs was monitored by confocal microscopy of *in situ* fixed cell cultures. Optimal test conditions were determined (Fig.3_S1A-F) and the results are summarized in **Fig.3I**. Original LASX files which served as a basis of our quantification, are publicly available (doi: 10.6019/S-BIAD1456). An example for our approach of counting the MV-IEVs is shown on Fig3_S1H. Cytochalasin B did not have any effect on the discharge of MV-IEVs suggesting that the release did not involve a major actin-dependent mechanism. In contrast, there was a significant reduction of the MV-IEV secretion upon exposure of the cells to Colchicine indicating a role of microtubules in the release of the MV-IEVs. While Rapamycin significantly reduced the discharge of MV-IEVs, Chloroquine and Bafilomycin induced an enhanced MV-IEV secretion. Rapamycin activates autophagic degradation¹⁷, therefore, it induces a shift towards degradation as opposed to secretion. The lysosomotropic agents Chloroquine and Bafilomycin are known to interfere with the acidification of lysosomes^{18,19}. By blocking the degradation pathway of MVBs/amphisomes (**Fig.3H**), an enhanced sEV secretion is observed. This effect is well known for exosome secretion from MVBs^{20,21}. The diameters of the released MV-IEVs were determined based on confocal images (Fig.3_S1G). Metabolic activity of the cells was determined by a Resazurin assay, and a significant reduction was detected upon exposure of the cells to Rapamycin (Fig.3_S1D) in line with previously published data²². LC3II is the membrane-associated, lipidated autophagic form of LC3²³ and it is the hallmark of autophagy related membranes¹⁶. Importantly, by Western blot, we not only showed the presence of the membrane-bound LC3II in serum-free, IEV-depleted (sEV containing) conditioned

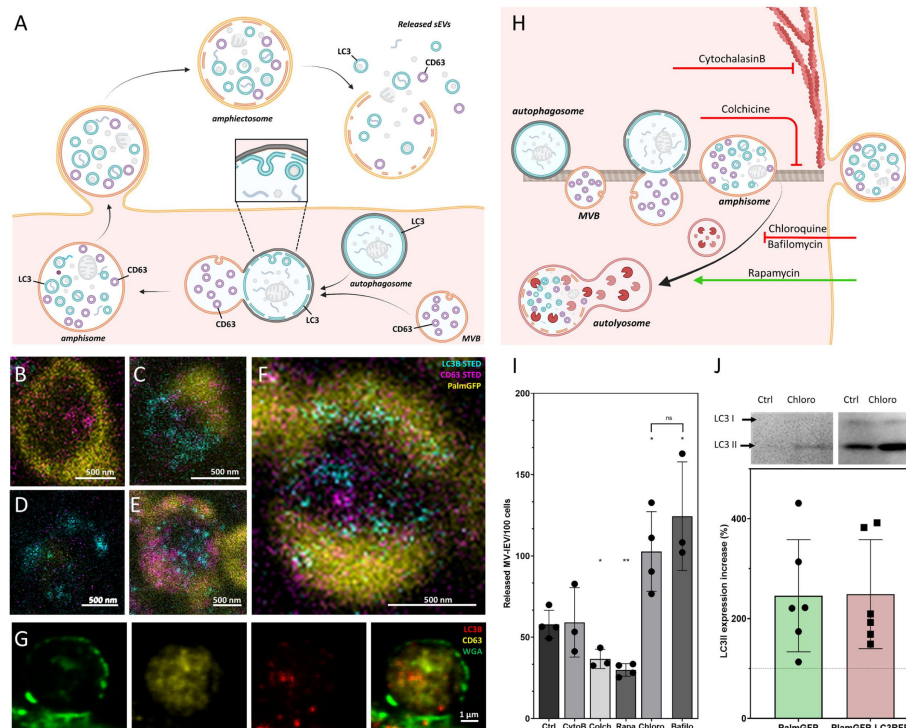


Figure 3.

Amphiectosome release and its modulation

Based on our data, a model of amphiectosome release was generated (A). According to this model, the fusion of MVBs and autophagosomes forms amphisomes. The LC3B positive membrane layer (indicated in cyan) undergoes disintegration and forms LC3B positive ILVs inside the amphisome. Later, the amphisome is released into the extracellular space by ectocytosis and can be identified extracellularly as an amphiectosome. Finally, the limiting membrane(s) of the amphiectosome is ruptured and the ILVs are released as sEVs into the extracellular space by a “torn bag mechanism”. Steps of amphisome formation including LC3 positive ILV formation in 30 μ M Chloroquine-treated HEK293T-PalmGFP cells was followed by super-resolution (STED) microscopy (B-F). The super-resolution STED channels were LC3B (cyan) and CD63 (magenta), while yellow indicates the confocal PalmGFP signal. Intracellular vesicular structures (such as endosomes, MVBs and amphisomes) may receive Palm-GFP from the plasma membrane. An MVB (B), an autophagosome with Palm-GFP negative membrane (D), fusion of an autophagosome and an MVB (C), formation of LC3B positive ILVs in an amphisome (F) and a mature amphisome (E) were detected. To confirm the origin of the external membrane layer of amphiectosomes, fluorescently labelled WGA was applied. The plasma membrane of the living non-fluorescent HEK293 cells was labelled. As the external membrane of the budding amphiectosome was WGA positive, its plasma membrane origin is confirmed (G). In order to further support our model on amphiectosome release and “torn bag” EV secretion, different *in vitro* treatments were applied. Cytochalasin B, Colchicine, Chloroquine, Bafilomycin A1 and Rapamycin were used to modulate amphiectosome release. Targeted molecular processes are summarized (H). While Cytochalasin B inhibits actin-dependent membrane budding and cell migration, Colchicine blocks the microtubule-dependent intracellular trafficking. While Chloroquine and Bafilomycin have similar, Rapamycin has opposite effect on lysosome-autophagosome or lysosome-amphisome fusion. Chloroquine and Bafilomycin inhibit lysosomal degradation while Rapamycin accelerates it. Based on confocal microscopy, Cytochalasin B (CytoB) did not alter the dynamics of amphiectosome release (I). In contrast, both Colchicine (Colch) and Rapamycin (Rapa) significantly inhibited the release of amphiectosomes, while Chloroquine (Chloro) and Bafilomycin (Bafilo) increased the release frequency. There was no difference between the effect of Chloroquine and Bafilomycin (I). Results are shown as mean $\pm$ SD of 3-4 independent biological replicates, analyzed by one-way ANOVA and Student’s t test, *: $p < 0.05$, **: $p < 0.01$, ns: non-significant. Original LASX files, which served as a basis of our quantification, are publicly available (doi: 10.6019/S-BIAD1456). Example for the calculation is shown on Fig3_S1H. Presence of membrane-bound (lipidated) LC3II was tested by Western blotting. The total protein content of serum-, cell- and large EV-depleted conditioned medium of HEK293T-PalmGFP (PalmGFP) and HEK293T-PalmGFP-LC3RFP (PalmGFP-LC3RFP) cells was precipitated by TCA and 20 μ g of the protein samples were loaded on the gel (J). The lipidated LC3II band was detected in all cases. Relative expression of control (Ctrl) and Chloroquine (Chloro)-treated samples were determined by densitometry. Chloroquine treatment increased the LC3II level by approximately two fold. Results are shown as mean $\pm$ SD of $n = 6$ biological replicates.

medium of both HEK293T-PalmGFP and HEK293T-PalmGFP-LC3RFP cells, but the amount of LC3II substantially increased upon Chloroquine treatment (**Fig.3J**). Raw data of Western blots are available in Fig.3_S2.

Recent advances in the EV field shed light on migrasomes, a special type of MV-IEVs^{24,25}. With their pomegranate-like ultrastructure, migrasomes resemble amphiectosomes. Therefore, we tested the presence of TSPAN4, a migrasome limiting membrane marker²⁵, in amphiectosomes. **Fig.4A,B,G,H** show that although TSPAN4 was present intraluminally in the HEK293T-PalmGFP-derived MV-IEVs, it was clearly absent from their external membrane. Surprisingly, we identified two different HT29 cell-derived MV-IEV populations: one in which TSPAN4 was only located intraluminally (**Fig.4C,E,I,K**), and another one with a TSPAN4 positive external membrane (**Fig.4D,F,J,L**). This raised the possibility that the latter population corresponded to migrasomes. Our co-localization analysis also confirmed the existence of two distinct MV-IEV populations (**Fig.4M**). Next, we carried out live-cell imaging on HEK293T-PalmGFP-LC3RFP cells. The released MV-IEVs were either LC3 positive or negative intraluminally (**Fig.4N,O**). Our TEM images confirmed that certain cell types can release both migrasome-like structures and amphiectosomes. MV-IEVs with typical migrasome-associated retraction fiber(s) were detected in the case of HL1 (**Fig.4P**), HEK293T-PalmGFP (**Fig.4Q**) and BMMC cells (**Fig.4R**). Of note, it cannot be excluded that the elongated structures observed in the above cases, may correspond to tunnelling nanotubes²⁶. Importantly, the same cell lines also released amphiectosomes by budding from the cell surface (**Fig.4S-U**). Taken together, based on the absence of TSPAN4 in their external membrane, and their lack of association with retraction fibers, amphiectosomes appear to be distinct from migrasomes. Besides migrasomes, another MV-IEV type was described in the case of gastrointestinal tumors and low-grade glioblastoma cells referred to as spheresome^{27,28}. However, there is no data on a relationship of spheresome release and autophagy. Recently, endothelial cell derived, multicompartmented microvesicles (MCMVs) were shown to protrude and pinch-off from the cell surface releasing ILVs by a mechanism similar to exocytosis²⁹. The absence of protrusion clusters described for MCMV²⁹ distinguishes amphiectosomes from MCMVs. In addition, the previously described so called “nodal vesicular parcels”³⁰ might be special examples of amphiectosomes. Finally, in *C. elegans*, the release autophagy and stress related large EVs (exophers) has been documented^{31,32,33}. They contain damaged organelles and do not have a MV-IEV like ultrastructure. In contrast, the amphiectosomes we described here, have multivesicular structure without recognizable damaged organelles.

Our approach, involving *in situ* fixation of cultures and tissues, made it possible to recognize sEV release from amphiectosomes by the “torn bag mechanism”. We propose that this mechanism could be easily missed earlier if conditioned medium was subjected to centrifugation, SEC purification or even to simple pipetting, which may rupture the limiting membrane of amphiectosomes. This aligns with our observation that the spontaneous escape of ILVs from untouched amphiectosomes can be completed as early as 5 minutes after amphiectosome release. Based on our data presented here, and considering that the exocytosis of MVBs/amphisomes under steady-state conditions is rarely documented in the scientific literature, we suggest that amphiectosome secretion and the “torn bag mechanism” may have a significant, yet previously unrecognized, role in sEV biogenesis.

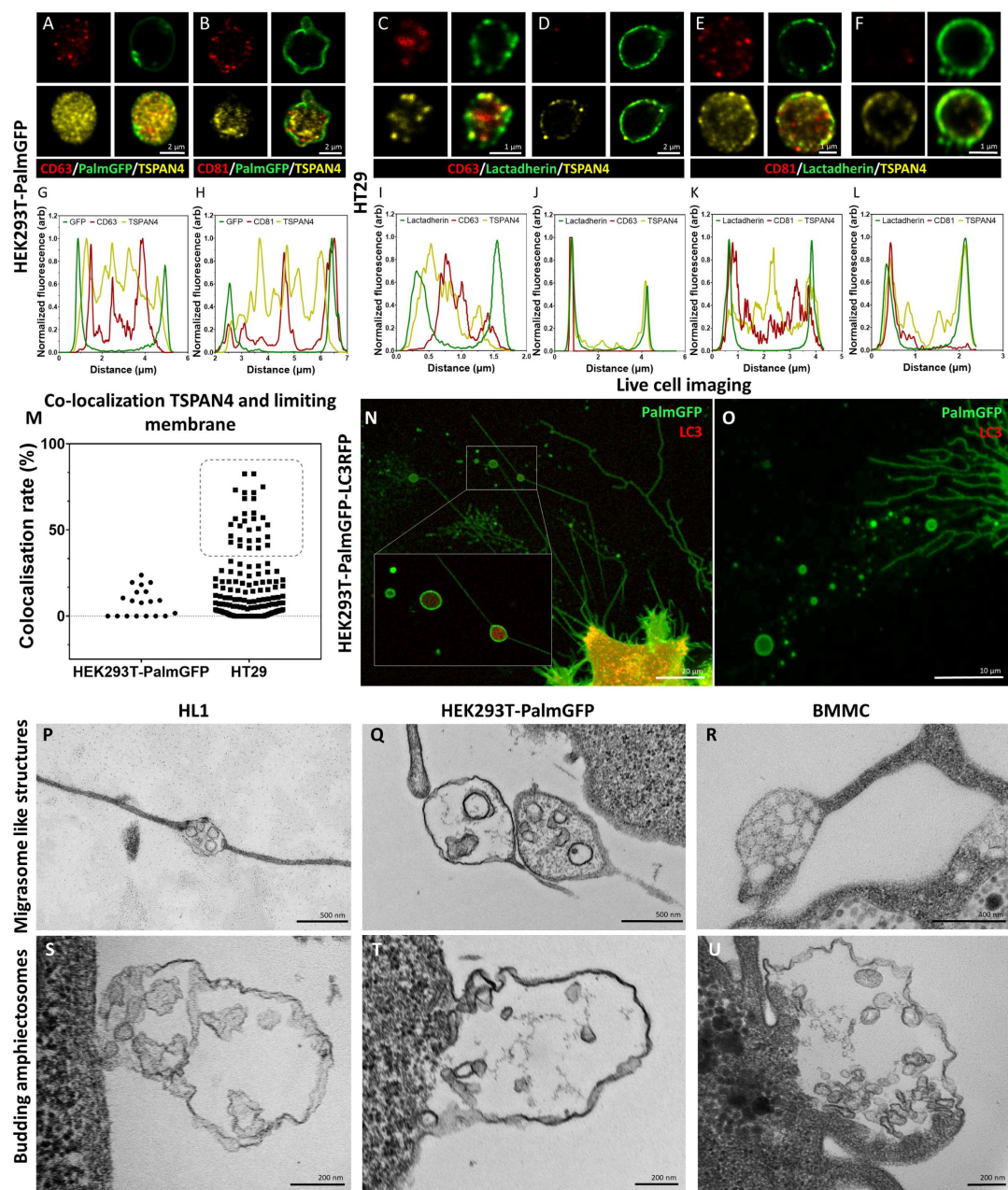


Figure 4.

Comparison of amphiectosomes and migrasomes

Commonly used sEV markers (CD63, CD81) and TSPAN4, a suggested migrasome marker, were tested in *in situ* fixed intact MV-IEVs of HEK293T-PalmGFP (A,B) and HT29 (C-F) cells by confocal microscopy. Normalized fluorescence intensities were calculated in order to determine the relative localization of the limiting membrane (with PalmGFP or lactadherin labelling) and the CD63/TSPAN4 and CD81/TSPAN4 markers (G-L). In the case of HEK293T-PalmGFP-derived EVs, we did not find migrasomes with TSPAN4 in their limiting membrane. The TSPAN4 signal was only detected intraluminally in the MV-IEVs. The limiting membranes of HT29-derived MV-IEVs were either TSPAN4 positive and negative. The co-localization rate between the limiting membrane and TSPAN4 was low in case of HEK293T-PalmGFP-derived EVs. In the case of HT29 cells, two MV-IEV populations were identified: one with low and one with high co-localization rates (M). Live cell imaging of HEK293T-PalmGFP-LC3RFP cells showed retraction fiber-associated MV-IEVs with or without intraluminal LC3 positivity (N,O). Using TEM, we could identify structures with retraction fiber-associated morphology in the case of HL1 (P), HEK293T-PalmGFP (Q) and BMMC (R) cells. For comparison, budding of amphiectosomes of the same HL1 (S), HEK293TPalmGFP (T) and BMMC cells (U) are shown (without being associated with long retractions fibers).

Material and methods

Key Resources Table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (<i>Homo sapiens</i>)	HEK293 human kidney(embryonic)	ECACC (Sigma)	#85120602	Batch No: 18E026
Cell line (<i>Homo sapiens</i>)	HT29 Caucasian colon adenocarcinoma grade II	ECACC (Sigma)	#91072201	Batch No 09K003
Cell line (<i>Homo sapiens</i>)	HepG2 human hepatocyte carcinoma	ECACC (Sigma)	#85011430	Batch No: 19B009
Cell line (<i>Mus musculus</i>)	HL1 mouse cardiomyocyte cell line, atrial	Merck	# SCC065	Batch No: RD1601001
Cell line (<i>Homo sapiens</i>)	HEK293TPalmGFP human kidney(embryonic) expressing palmitoylated GFP	Kind gift of Charles Lai https://doi.org/10.1038/ncomms8029		Resorted before MCB preparation
Cell line (<i>Homo sapiens</i>)	HEK293T- PalmGFPLC3RFP human kidney(embryonic) expressing palmitoylatedGFP and RFP tagged LC3	This paper		See material and methods
Cell line (<i>Mus musculus</i>)	BMMC bone marrow derived mast cells	Primary cell culture https://doi.org/10.1002/jev.2.12023		
Lentiviral particles	LentiBrite RFP-LC3 Lentiviral Biosensor	Merck	17-10143	Batch No: 3530171

Biological sample (Mus musculus)	own animal house		C57BL/6	male, 12 weeks old
Antibody	rabbit polyclonal anti-CD63 (Cterminal)	Sigma/Merck	SAB2109138	IF (1:200) WB (1:500)
Antibody	mouse monoclonal anti-CD63	SantaCruz Biotechnology	MX-49.129.5 clone: sc-5275	IF (1:200) TEM (1:50) WB (1:1,000)
Antibody	rabbit polyclonal anti-CD81	Sigma/Merck	SAB3500454	IF (1:200) WB (1:2,500)
Antibody	mouse monoclonal anti-CD81	Invitrogen	MA5-13548 clone: 1.3.3.22	IF (1:100) WB (1:100)
Antibody	rabbit polyclonal anti-TSG101	Sigma/Merck	HPA006161	IF (1:200) WB (1:1,000)
Antibody	rabbit polyclonal anti-ALIX (Cterminal)	Sigma/Merck	SAB420047	IF (1:200) WB (1:1,000)
Antibody	rabbit monoclonal anti-LC3B	Sigma/Merck	ZRB100 clone: 12K5	IF (1:200) TEM (1:50) WB (1:1,000)
Antibody	rabbit monoclonal anti-LC3A	Sigma/Merck	ZRB1125 clone: 3J12	IF (1:200) WB (1:1,000)
Antibody	rabbit polyclonal anti-TSPAN4	Bioss	BS-9413R	IF (1:200)
Antibody	mouse monoclonal anti-Rab7	Sigma/Merck	R8779 clone: Rab7117	IF (1:200) WB (1:1,000)

Antibody	mouse monoclonal anti- α -tubulin	Sigma/Merck	T9026 clone: DM1A	IF (1:200)
Antibody	mouse monoclonal anti-GFP	Sigma/Merck	G6539 clone: GFP-20	IF (1:200) WB (1:1,000)
Antibody	mouse monoclonal anti-RFP	Invitrogen	MA5-15257 clone: RF5R	WB (1:1,000)
Antibody	goat antimouse IgGATTO550	Sigma/Merck	43394	IF (1:1,000)
Antibody	goat antirabbit IgG-ATTO647N	Sigma/Merck	40839	IF (1:1,000)
Antibody	goat antimouse Star 635P	Abberior	ST635P-1001-500UG	IF (1:500)
Antibody	goat antirabbit Star 580	Abberior	ST580-1002-500UG	IF (1:500)
Antibody	goat polyclonal anti-rabbit IgG Fc (HRP)	abcam	ab97200	WB (1:10,000)
Antibody	goat polyclonal anti-mouse IgG Fc (HRP)	abcam	ab97265	WB (1:10,000)
Antibody	goat polyclonal anti-rabbit IgG (whole molecule) 10 nm gold preadsorbed	abcam	ab27234	TEM (1:50)
Antibody	goat polyclonal anti-mouse IgG (whole molecule) 5 nm gold preadsorbed	Sigma/Merck	G7527	TEM (1:50)

Special reagent	CF488A conjugated Wheat Germ Agglutinin (WGA)	Biotium	29022-1	Lot Number: 21C0224-1149057
Chemical compound, drug	Bafilomycin A1	Sigma/Merck	B1793	Lot Number: 0000190389
Chemical compound, drug	Colchicine	Serva	77120.02	Lot Number: 190300
Chemical compound, drug	Chloroquinediphosphate	Invitrogen	P36236 C	Lot Number: 2441325
Chemical compound, drug	Rapamycin	Sigma/Merck	R0395	Lot Number: 0000084976
Chemical compound, drug	Cytochalasin B	Sigma	C2743	Lot Number: 037M4083V
Chemical compound, drug	FBS	Biosera	FB-1090/500	Lot Number: 015BS575
Special equipment	TFF Easy	HansaBioMed Life Sciences	HBM-TFF/1	
Software, algorithm	LASX	Leica	Leica Application Suite X 3.5.5.19976	
Software, algorithm	ZEN Blue	Zeiss	ZEN 2.3 lite	
Software, algorithm	iTEM	Olympus	iTEM 5.1	

Software, algorithm	Image J	https://imagej.net/ij/	v1.54g	
Software, algorithm	Prism9	GraphPad	GraphPad Prism 9.4.1	
Software, algorithm	Biorender	https://www.biorender.com/		

Cell lines

The HEK293 human embryonic kidney, the HepG2 human hepatocyte carcinoma cell line, the HT29 human colon adenocarcinoma cell lines and the H9c2 rat cardiomyoblast cell line were purchased from the European Collection of Authenticated Cell Cultures (ECACC) through their distributor (Sigma). The HL1 cell line was purchased from Millipore. The HEK293TPalmGFP human embryonic kidney cells was kindly provided by Charles P. Lai³⁴. Mouse bone marrow-derived mast cells (BMMCs) were differentiated and expanded as we described previously³⁵. The HEK293, HEK293TPalmGFP, HepG2 and H9c2 cell lines were grown in DMEM (Gibco)^{36–38}, the HT29 cells were cultured in RPMI 1640 (Gibco)¹², while the HL1 cells were grown in Claycomb medium³⁶. All cells were cultured with 10 % fetal bovine serum (FBS, BioSera) in the presence of 100 U/mL of Penicillin and 100 µg/mL Streptomycin (Sigma). Before analysis by confocal microscopy, the cells were cultured on the surface of gelatin-fibronectin coated glass coverslips (VWR). The coating solution contained 0.02 % gelatin (Sigma) and 5 mg/mL fibronectin (Invitrogen). Coverslips were coated overnight (O/N) at 37 °C.

For transmission electron microscopy (TEM), the adherent cells (HEK293, HEK293T-PalmGFP, HepG2, HT29 and HL1) were grown in gelatin-fibronectin coated 8-well Flux Cell Culture Slides (SPL).

Cell cultures were tested regularly for Mycoplasma infection by PCR, with the following PCR primers: GAAGAWATGCCWTATTTAGAAGATGG and CCRTTTTGACTYTTWCCAC-CMAGTGGTTGTTG³⁶.

Generation of HEK293T-PalmGFP-LC3RFP cell line

For generation of a stable HEK293T-PalmGFP-LC3RFP cell line, HEK293T-PalmGFP cells were transfected by LentiBrite RFP-LC3 Lentiviral particles (Merck) according to the instructions of the manufacturer. The GFP-RFP double positive cells were sorted by a HS800 Cell Sorter (SONY) and cell banks (master and working cell banks) were prepared. The success of the stable transfection was analyzed by immunocytochemistry and Western blotting. Results are shown in Fig.2_S5.

Confocal microscopy

Confocal microscopy was carried out as we described earlier³⁶ with some modifications. As serum starvation significantly affects autophagy³⁹, and EV-depleted FBS in the cell culture medium may influence cellular physiology and morphology⁴⁰, FBS was not removed before fixation. Our study focuses on large EV (IEVs) with diameter > 350-500 nm. EVs in this size range, are neglectible in FBS because of sterile filtration and heat inactivation of the serum. Unlike the majority of the studies in the field of EVs, here we analyzed untouched, *in situ* fixed and cultured cells together with their microenvironment. Since centrifugation may disrupt the limiting membrane of amphiectosomes, the *in situ* fixation made it possible to observe them in their intact form. The culture medium was gently removed by pipetting from above the cells leaving a thin medium layer only (approximately 150 µL of liquid on the cells). Without any further washing, cells were *in situ* fixed by 4 % paraformaldehyde (PFA) in phosphate buffer saline (PBS) for 20 min at room temperature (RT). The released IEVs were either fixed and captured during the release or were preserved on the gelatin/fibronectin surface coating. After fixation, 3x 5 min washes with 50 mM glycine in PBS were carried out. In the case of the non-fluorescent HepG2 and HT29 cells, a lactadherin-based plasma membrane staining was performed^{14,35,37}. Lactadherin (Haematologic Technologies) was conjugated to ATTO488 fluorophore (abcam) according to the instructions of the manufacturer. The lactadherin-ATTO488 conjugate was added to the fixed cells in 1:100 dilution in PBS (for 1 h, RT) before permeabilisation. The unbound lactadherin was removed by washing with PBS (3 times, 5 min, RT) and post-fixation was carried out by 4 % PFA (20 min, RT). PFA was removed by washes with 50 mM glycine in PBS (3 times, 5 min, RT). Blocking

and permeabilization of the cells were performed by 10 % FBS with 0.1 % TritonX-100 (Sigma) in PBS (1 h, RT). In general, primary antibodies were applied in 1:200 dilution overnight (O/N) at 4 °C in the above blocking and permeabilization solution. Excess primary antibodies were eliminated by washing with the blocking and permeabilization solution (3 times, 5 min RT). The secondary antibodies were applied in 1:1000 dilution in 1 % FBS in PBS (1 h, RT). Unbound secondary antibodies were eliminated by washing (1 % FBS, in PBS, 2 times, 5 min; PBS 2 times, 5 min; water 2 times, 5 min) and the samples were mounted in ProLong Diamond with DAPI (Invitrogen).

In order to provide evidence for the plasma membrane origin of the outer membrane layer of amphictosomes, HEK293 were cultured on glass cover slips (VWR). Reaching 60 % confluency, the cells were incubated in expansion medium with 5 µg/mL CF488A conjugated Wheat Germ Agglutinin (WGA) for 30 min at 37°C. After labelling the surface of the plasma membrane by WGA, cells were washed three times by expansion medium and were cultured for an additional 3 hours. Next, they were fixed by 4 % PFA (20 min, RT). LC3B and CD63 labelling were performed as described above.

Microscopic slides were examined by Leica SP8 Lightning confocal microscope with adaptive lightning mode using a HC PL APO CS2 63x/1.40 OIL objective with hybrid detector. Where we showed released MV-IEVs, they were not joined to cells in another detected Z-plane. The applied lookup tables (LUT) were linear during this study. For image analysis and co-localization studies, we applied Leica LASX software using the unprocessed raw images. In case of co-localization studies, 20 % threshold and 10 % background settings were applied.

Multi-channel STED super-resolution imaging

Immunofluorescent labeling for multi-channel stimulated emission depletion (STED) nanoscopy was performed as in the case of confocal microscopy. The used primary antibodies were: LC3B (rabbit) and CD63 (mouse). Abberior Star 635P Goat anti Mouse Abberior Star 580 Goat anti Rabbit secondary antibodies for STED microscopy have been obtained from Abberior GmbH. Samples were mounted with SlowFade™ Diamond Antifade Mountant (Thermo). Immunofluorescence was analyzed using an Abberior Instruments Facility Line STED Microscope system built on an Olympus IX83 fully motorized inverted microscope base (Olympus), equipped with a ZDC-830 TrueFocus Z-drift compensator system, an IX3-SSU ultrasonic stage, a QUADScan Beam Scanner scanning head, APD detectors and an UPLXAPO60XO 60X oil immersion objective (NA 1.42). We used the 488, 561 and 640 nm solid state lasers for imaging, and a 775 nm solid state laser for STED depletion. Image acquisition was performed using the Inspector data acquisition software (version: 16.3.14278-w2129-win64).

Purification of sEV fraction

Small EV fractions for TEM analysis were separated from serum-free conditioned medium of HEK293T-PalmGFP cells by gravity-filtration, differential centrifugation and tangential flow filtration (TFF Easy, HansaBiomed) as described previously³⁸.

Transmission electron microscopy

Adherent cells (HEK293, HEK293T-PalmGFP, HepG2, HT29 and HL1), as well as mouse (C57BL/6) kidney and liver tissues pieces (approx. 1.5 mm x 1.5 mm) were immersed in and fixed by 4 % glutaraldehyde (48 h, 4 °C), post-fixed by 1 % osmium tetroxide (2 h, RT) and were embedded into EPON resin (Electron Microscopy Sciences) as described previously⁴¹. In the case of BMMCs, 920 µL cell suspension was complemented with 80 µL 50 % glutaraldehyde to reach the final 4 % glutaraldehyde concentration. Cells were fixed for 48 h at 4 °C and were post-fixed by 1% osmium tetroxide (2 h, RT). During sample preparation, BMMCs were collected by gravity-based

sedimentation. Due to the high viscosity of EPON resin, BMNCs were embedded in LR White low viscosity resin (SPI Supplies) according to the instructions of the manufacturer. Ultrathin sections (60 nm) were contrasted by uranyl acetate (3.75 %, 10 min, RT) and lead citrate (12 min, RT).

For immunogold labelling of ultrathin sections, cells and tissues were fixed by 4 % PFA with 0.1 % glutaraldehyde (48 h, 4 °C) and were post-fixed by 0.5 % osmium tetroxide (30 min, RT). Samples were embedded into LR White hydrophilic resin. The sections were exposed to H₂O₂ and NaBH₄ to render the epitopes accessible and were immunogold labelled as described previously¹². The contrast was enhanced by uranyl acetate (3.75 %, 1 min, RT) and lead citrate (2 min, RT).

HEK293T-PalmGFP-derived sEVs separated from serum-free conditioned medium, were detected by negative-positive contrasting without embedding and sectioning¹⁵. Immunogold labelling was performed as described previously³⁶. Antibodies were used in 1:50 dilution.

Detailed list of the used antibodies is available in the Key Resources Table.

For all electron microscopic studies, a JEOL 1011 transmission electron microscope was used. Images were captured with the help of Olympus iTEM software and for image analysis, Image J software was used.

Live cell imaging

The HEK293T-PalmGFP-LC3RFP stable cell line was cultured the same way as HEK293T-PalmGFP cells. Before the experiments, gelatin-fibronectin coated 10-well coverslip bottom chamber slide (Greiner-BioOne) was seeded and treated by 30 µM Chloroquine O/N. Release of migrasomes, amphiectosomes and sEVs were followed by the Leica SP8 Lightning confocal microscope equipped with an Okolab environmental chamber and a Zeiss ELYRA 7 with Lattice SIM² super-resolution fluorescent microscope with the help of 63x/1.4 planapochromat Oil objective. For image analysis, we applied Leica LASX, Zeiss ZEN Blue and Image J softwares.

Modulation of amphiectosome release

In order to test the release mechanism of amphiectosomes and to distinguish them from migrasomes, different treatments were applied overnight in fresh, serum containing cell culture medium except for Colchicine, where 1 h treatment was selected. Maturation and fusion of endosomes and lysosomes were inhibited by 30 µM Chloroquine (Invitrogen) or 10 nM BafilomycinA1 (Sigma). Actin polymerization was inhibited by 125 ng/mL Cytochalasin B (Sigma). Tubulin polymerization and function were inhibited by 250 pg/mL Colchicine, while an autophagy-related degradation was induced by 50 ng/mL Rapamycin. The selected concentrations were determined based both on literature data and our preliminary experiments (Fig.3_S1). Cellular metabolic activity was determined by a metabolic activity-based Resazurin assay³⁶. Fresh cell culture medium was added to control cultures a day before the *in situ* fixation. Reagents were diluted in fresh cell culture medium. Leica TCS SP8 Lightning confocal microscope was used for detection of amphiectosome release. A few hundred µm² sized area with 15-20 µm in height was tile-scanned with a few hundred cells. The MV-IEVs were recognized as CD63 positive EVs surrounded by GFP positive membrane. They were counted and were normalized to the number of nuclei. Raw images were deposited in BioImage Archive (<https://www.ebi.ac.uk/bioimage-archive/>) with the accession number S-BIAD1456 (doi: 10.6019/S-BIAD1456).

Western blotting

Presence of proteins and specificity of the used primary antibodies were confirmed by Western blotting as described previously³⁶. For accurate quantification (free from variations potentially caused by EV purification), we analyzed cell-, serum- and large EV (diameter > 800 nm) free conditioned medium. The cells were cultured O/N in a serum-free culture medium. After

harvesting cells were eliminated by centrifugation (300 g, 10 min at 4 °C) followed by a 2.000 g centrifugation (30 min at 4 °C) to eliminate IEVs. Total protein content of the conditioned, serum-, cell- and IEV-free medium was precipitated by trichloroacetic acid as described previously^{36,42}. The protein pellets were suspended in cOmplete Protease Inhibitor Cocktail (Roche) containing radio-immunoprecipitation assay (RIPA) buffer.

When whole cell lysate was tested for validation of antibodies and the HEK293T-PalmGFP-LC3RFP cell line, cells were lysed in cOmplete Protease Inhibitor Cocktail (Roche) containing RIPA buffer.

Polyacrylamide gel electrophoresis was carried out using 10 % gels (acrylamide/bis-acrylamide ratio 37.5:1) or any kDa precast gels (Biorad) and a MiniProtean (BioRad) gel running system. For better solubilization of membrane proteins, equal volumes of 0.1 % TritonX-100, Laemmli buffer and samples were mixed as described previously⁴³. Approximately 10-30 µg protein were loaded into each well. Following electrophoretic separation, proteins were transferred to PVDF membranes (Serva). Membranes were blocked with 5 % skimmed milk powder or 5 % BSA in washing buffer for 1 h. Primary antibodies were applied in 1:1000 dilution with the exception of the anti-CD63 (rabbit), anti-CD81 (rabbit) and anti-CD81 (mouse) antibodies where 1:500, 1:2500 and 1:100 dilutions were used, respectively. Peroxidase-labelled secondary antibodies were applied in 1:10 000 dilution. The signals were detected by ECL Western Blotting Substrate (Thermo Scientific) with an Imager CHEMI Premium (VWR) image analyzer system. In case of quantification, equal protein amounts were loaded to the gels. Within a biological replicate, the control and Chloroquine-treated samples were run on the same gels. To enable comparison, the relative expression of control and Chloroquine-treated samples were determined and compared.

Software and statistical analysis

For image capturing, analysis and co-localization studies, Leica LAS X, Zeiss ZEN Blue, Olympus iTEM and Image J software were used. Figures and graphs were generated using GraphPad Prism 9.4.1 and *BioRender.com* (Figure 1G, Figure 1N, Figure 1U, Figure 3A, Figure 3H, Figure 2 - Figure supplement 6A, Figure 2 - Figure supplement 6B, Figure 2 - Figure supplement 6C, Figure 2 - Figure supplement 6D, Figure 2 - Figure supplement 6E). For statistical analysis, standard deviation was calculated. Unpaired two tailed Student's t-tests and one-way ANOVA were used (* p<0.05, ** p<0.01, *** p<0.001, **** p<0.0001).

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Supplementary materials

Supplementary Material [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors' research group had previously demonstrated the release of large multivesicular body-like structures by human colorectal cancer cells. This manuscript expands on their findings, revealing that this phenomenon is not exclusive to colorectal cancer cells but is also observed in various other cell types, including different cultured cell lines, as well as cells in the mouse kidney and liver. Furthermore, the authors argue that these large multivesicular body-like structures originate from intracellular amphisomes, which they term "amphictosomes." These amphictosomes release their intraluminal vesicles (ILVs) through a "torn-bag mechanism." Finally, the authors demonstrate that the ILVs of amphictosomes are either LC3B positive or CD63 positive. This distinction implicates that the ILVs either originate from amphisomes or multivesicular bodies, respectively.

Strengths:

The manuscript reports a potential origin of extracellular vesicle (EV) biogenesis. The reported observations are intriguing.

Weaknesses:

In their revised version, the authors have addressed the majority of my criticisms. I have no further concerns regarding this manuscript.

<https://doi.org/10.7554/eLife.95828.2.sa2>

Reviewer #2 (Public review):

Summary:

authors had previously identified that a colorectal cancer cell line generates small extracellular vesicles (sEVs) via a mechanism where a larger intracellular compartment containing these sEVs is secreted from the surface of the cell and then tears to release its contents. Previous studies had suggested that intraluminal vesicles (ILVs) inside endosomal multivesicular bodies and amphisomes can be secreted by fusion of the compartment with the plasma membrane. The 'torn bag mechanism' considered in this manuscript is distinctly different, because it involves initial budding off of a plasma membrane-enclosed compartment (called the amphictosome in this manuscript, or MV-IEV). The authors successfully set out to investigate whether this mechanism is common to many cell types and to determine some of the subcellular processes involved.

The strengths of the study are:

- (1) The high-quality imaging approaches used, including live-cell imaging and EN, which seem to show good examples of the proposed mechanism.
- (2) They screen several cell lines for these structures, also search for similar structures in vivo, and show the tearing process by real-time imaging.
- (3) Regarding the intracellular mechanisms of ILV production, the authors also try to demonstrate the different stages of amphictosome production and differently labelled ILVs using immuno-EM.

Several of the techniques employed are technically challenging to do well, and so these are critical strengths of the manuscript.

Overall, I think the authors have been successful in identifying amphiectosomes secreted from multiple cell lines and cells in vivo, and in demonstrating that the ILVs inside them have at least two origins (autophagosome membrane and late endosomal multivesicular body) based on the markers that they carry. Inevitably, it remains unclear how universal this mechanism is in vivo and its overall contribution to EV function.

I think there could be a significant impact on the EV field and consequently on our understanding of cell-cell signalling based on these findings. It will flag the importance of investigating the release of amphiectosomes in other studies, especially as the molecular mechanisms involved in this type of 'ectosomal-style' release will be different from multivesicular compartment fusion to the plasma membrane and should be possible to be manipulated independently.

In general, the EV field has struggled to link up analysis of the subcellular biology of sEV secretion and the biochemical/physical analysis of the sEVs themselves, so from that perspective, the manuscript provides a novel angle on this problem.

<https://doi.org/10.7554/eLife.95828.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors' research group had previously demonstrated the release of large multivesicular body-like structures by human colorectal cancer cells. This manuscript expands on their findings, revealing that this phenomenon is not exclusive to colorectal cancer cells but is also observed in various other cell types, including different cultured cell lines, as well as cells in the mouse kidney and liver. Furthermore, the authors argue that these large multivesicular body-like structures originate from intracellular amphisomes, which they term "amphiectosomes." These amphiectosomes release their intraluminal vesicles (ILVs) through a "torn-bag mechanism." Finally, the authors demonstrate that the ILVs of amphiectosomes are either LC3B positive or CD63 positive. This distinction implies that the ILVs either originate from amphisomes or multivesicular bodies, respectively.

Strengths:

The manuscript reports a potential origin of extracellular vesicle (EV) biogenesis. The reported observations are intriguing.

Weaknesses:

It is essential to note that the manuscript has issues with experimental designs and lacks consistency in the presented data. Here is a list of the major concerns:

(1) The authors culture the cells in the presence of fetal bovine serum (FBS) in the culture medium. Given that FBS contains a substantial amount of EVs, this raises a significant issue, as it becomes challenging to differentiate between EVs derived from FBS and those released by the cells. This concern extends to all transmission electron microscopy (TEM)

images (Figure 1, 2P-S, S5, Figure 4 P-U) and the quantification of EV numbers in Figure 3. The authors need to use an FBS-free cell culture medium.

Although FBS indeed contains bovine EVs, however, the presence of very large multivesicular EVs (amphictosomes) that our manuscript focuses on has never been observed and reported. For reported size distributions of EVs in FBS, please find a few relevant references below:

PMID: 29410778, PMID: 33532042, PMID: 30940830 and PMID: 37298194

All the above publications show that the number of lEVs > 350-500 nm is negligible in FBS. The average diameter of MV-lEVs (amphictosomes) described in our manuscript is around 1.00-1.50 micrometer.

Reviewer #1: *These papers evaluated the effectiveness of various methods to eliminate EVs from FBS, emphasizing the challenges associated with the presence of EVs in FBS. They also caution against using FBS in EV studies due to these issues. However, I did not find a clear indication regarding the size distributions of EVs in FBS in these papers.*

Please provide accurate reference supporting the claim that 'lEVs > 350-500 nm are negligible in FBS.' The papers cited by the authors do not address this specific point.

In the revised manuscript, we addressed the point that due to sterile filtering of FBS, it cannot contain large >0.22 μ m EVs

Our response to Reviewer #1 point 2. When we demonstrated the TEM of isolated EVs, we consistently used serum- free conditioned medium (Fig2 P-S, Fig2S5 J, O) as described previously (Németh et al 2021, PMID: 34665280).

Reviewer #1: *This is an important point that is not mentioned in the original main text, figure legend or method. Please address.*

We agree and we apologize for it. We added this information to the revised manuscript.

Our response to Reviewer #1 point 3. Our TEM images show cells captured in the process of budding and scission of large multivesicular EVs excluding the possibility that these structures could have originated from FBS.

Reviewer #1: *These images may also depict the engulfment of EVs in FBS. Hence, it is crucial to utilize EV-free or EV-depleted FBS.*

As we mentioned earlier, we added the information to the revised manuscript that sterile filtering of the FBS presumably removed particles >0.22 μ m EVs

Our response to Reviewer #1 point 4. In addition, in our confocal analysis, we studied Palm-GFP positive, cell-line derived MV-lEVs. Importantly, in these experiments, FBS-derived EVs are non-fluorescent, therefore, the distinction between GFP positive MV-lEVs and FBS-derived EVs was evident.

Reviewer #1: *I agree that these fluorescent-labeled assays conclusively indicate that the MV-lEVs are originating from the cells. However, the images of concerns are the non-fluorescent-labeled images in (Figure 1, 2P-S, S5, Figure 4 P-U and Figure 3). The MV-lEVs may derive from both the cells and FBS.*

Please see above our response to points 1-3.

Our response to Reviewer #1 point 5. In addition, culturing cells in FBS-free medium (serum starvation) significantly affects autophagy. Given that in our study, we focused on autophagy related amphiectosome secretion, we intentionally chose to use FBS supplemented medium.

Reviewer #1 *If this is a concern, the authors should use EV-depletive FBS.*

As we discussed above, sterile filtration of FBS removes particles >0.22 μm . In addition, based on our preliminary experiments, EV-depleted serum may effect cell physiology.

Our response to Reviewer #1 point 6. Even though the authors of this manuscript are not familiar with the technological details how FBS is processed before commercialization, it is reasonable to assume that the samples are subjected to sterile filtration (through a 0.22 micron filter) after which MV-IEVs cannot be present in the commercial FBS samples.

Reviewer #1 *This is a fair comment that needs to be included in the manuscript.*

As you suggested, this comment is now included in the revised manuscript

(2) The data presented in Figure 2 is not convincingly supportive of the authors' conclusion. The authors argue that "...CD81 was present in the plasma membrane-derived limiting membrane (Figures 2B, D, F), while CD63 was only found inside the MV-IEVs (Fig. 2A, C, E)." However, in Figure 2G, there is an observable CD63 signal in the limiting membrane (overlapping with the green signals), and in Figure 2J, CD81 also exhibits overlap with MV-IEVs.

Both CD63 and CD81 are tetraspanins known to be present both in the membrane of sEVs and in the plasma membrane of cells (for references, please see Uniprot subcellular location maps: https://www.uniprot.org/uniprotkb/P08962/entry#subcellular_location https://www.uniprot.org/uniprotkb/P60033/entry#subcellular_location). However, according the feedback of the reviewer, for clarity, we will delete the implicated sentence from the text.

Reviewer #1 *Please also justify the statement questioned in (3) as these arguments are interconnected.*

We hope you find our above responses to your comment acceptable.

(3) Following up on the previous concern, the authors argue that CD81 and CD63 are exclusively located on the limiting membrane and MV-IEVs, respectively (Figure 2-A-M). However, in lines 104-106, the authors conclude that "The simultaneous presence of CD63, CD81, TSG101, ALIX, and the autophagosome marker LC3B within the MV-IEVs..." This statement indicates that CD63 and CD81 co-localize to the MV-IEVs. The authors need to address this apparent discrepancy and provide an explanation.

There must be a misunderstanding because we did not claim or implicate in the text that "CD81 and CD63 are exclusively located on the limiting membrane and MV-IEVs". Here we studied co-localization of the above proteins in the case intraluminal vesicles (ILVs). In Fig 2. we did not show any analysis of limiting membrane co-localization.

Reviewer #1 *I have indicated that this statement is found in lines 104-106, where the authors argue, 'The simultaneous presence of CD63, CD81, TSG101, ALIX, and the autophagosome marker LC3B within the MV-IEVs...' If the authors acknowledge the inaccuracy of this statement, please provide a justification for this argument.*

For clarity, we modified the description of data shown in Fig2 in the revised manuscript.

(4) The specificity of the antibodies used in Figure 2 should be validated through knockout or knockdown experiments. Several of the antibodies used in this figure detect multiple bands on western blots, raising doubts about their specificity. Verification through additional experimental approaches is essential to ensure the reliability and accuracy of all the immunostaining data in this manuscript.

We will consider this suggestion during the revision of the manuscript.

Reviewer #1: Please do so.

We carefully considered the suggestion, but we realized that it was not feasible for us to perform gene silencing in the case of all our used antibodies before resubmission of our revised manuscript. However, we repeated the Western blot for mouse anti-CD81 (Invitrogen MAA5-13548) and replaced the previous Western blot by it in the revised manuscript (Fig.2-S4H)

(5) In Figures 2P-R, the morphology of the MV-IEVs does not resemble those shown in Figures 1-A, H, and D, indicating a notable inconsistency in the data.

EM images in Figure2 P-R show sEVs separated from serum-free conditioned media as opposed to MV-IEVs, which were *in situ* captured in fixed tissue cultures (Fig1). Therefore, the two EV populations necessarily have different size and structure. Furthermore, Fig. 1 shows images of ultrathin sections while in Figure 2P-R, we used a negative-positive contrasting of intact sEV-s without embedding and sectioning.

(6) There are no loading controls provided for any of the western blot data.

Not even the latest MISEV 2023 guidelines give recommendations for proper loading control for separated EVs in Western blot (MISEV 2023 , DOI: 10.1002/jev2.12404 PMID: 38326288). Here we applied our previously developed method (PMID: 37103858), which in our opinion, is the most reliable approach to be used for sEV Western blotting. For whole cell lysates, we used actin as loading control (Fig3-S2B).

Reviewer #1: The blots referenced here (Fig2-S3; Fig2-S4B; Fig3-S2B) were conducted using total cell lysates, not EV extracts. Only one blot in Fig3-S2B includes an actin control. All remaining blots should incorporate actin controls for consistency.

Fig2-S3 (corresponding to Fig2-S4 in the revised manuscript) only shows reactivity of the used antibodies. This Western blot is not intended to serve as a basis of any quantitative conclusions. Fig2-S4 (corresponding to Fig2-S5 in the revised manuscript) includes the actin control. Fig3-S2B shows the complete membrane, which was cut into 4 pieces, and the immune reactivity of different antibodies was tested. The actin band was included on the anti-LC3B blot. For clarity, we rephrased the figure legend.

Additionally, for Figures 2-S4B, the authors should run the samples from lanes i-iii in a single gel.

Please note that in Figure 2- S4B, we did run a single gel, and the blot was cut into 4 pieces, which were tested by anti-GFP, anti-RFP, anti-LC3A and anti-LC3B antibodies. Full Western blots are shown in Fig.3_S2 B, and lanes “1”, “2” and “3” correspond to “i”, “ii” and “iii” in Fig.2-S4, respectively.

Reviewer #1: In the original Figure 2- S4B, the blots were sectioned into 12 pieces. If lanes "i," "ii," and "iii" were run on the same blot, the authors are advised to eliminate the grids between these lanes.

Grids separating the lanes have been eliminated on Fig.2_S4 (now Fig.2_S5 in the revised manuscript).

(7) In Figure 2-S4, is there co-localization observed between LC3RFP (LC3A?) with other MV-IFV markers? How about LC3B? Does LC3B co-localize with other MV-IFV markers?

In Supplementary Figure 2-S4, we showed successful generation of HEK293T-PalmGFP-LC3RFP cell line. In this case we tested the cells, and not the released MV-IEVs. LC3A co-localized with the RFP signal as expected.

Reviewer #1: Does LC3RFP colocalize with MV-IFV markers in HEK293T-PalmGFP-LC3RFP cell line? This experiment aims to clarify the conclusion made in lines 104-106, where the authors assert that 'The concurrent existence of CD63, CD81, TSG101, ALIX, and the autophagosome marker LC3B within the MV-IEVs...'

In the case of PalmGFP-LC3RFP cells, LC3-RFP is overexpressed. Simultaneous assessment of this overexpressed protein with non-overexpressed, fluorescent antibody-detected molecules proved to be challenging because of spectral overlaps and inappropriate signal-noise ratios. Furthermore, in association with EVs, the number of antibody-detected molecules is substantially lower than in cells. Therefore, even though we tried, we could not successfully perform these experiments.

(8) The TEM images presented in Figure 2-S5, specifically F, G, H, and I, do not closely resemble the images in Figure 2-S5 K, L, M, N, and O. Despite this dissimilarity, the authors argue that these images depict the same structures. The authors should provide an explanation for this observed discrepancy to ensure clarity and consistency in the interpretation of the presented data.

As indicated in Material and Methods, Fig 2-S5 F, G, H and I are conventional TEM images fixed by 4% glutaraldehyde 1% OsO₄ 2h and embedded into Epon resin with a post contrasting of 3.75% uranyl acetate 10 min and 12 min lead citrate. Samples processed this way have very high structure preservation and better image quality, however, they are not suitable for immune detection. In contrast, Fig.2-S5 K,L,M,N shows immunogold labelling of in situ fixed samples. In this case we used milder fixation (4% PFA, 0.1% glutaraldehyde, postfixing by 0.5% OsO₄ 30 min) and LR-White hydrophilic resin embedding. This special resin enables immunogold TEM analysis. The sections were exposed to H₂O₂ and NaBH₄ to render the epitopes accessible in the resin. Because of the different applied techniques, the preservation of the structure is not the same. In the case of Fig.2 J, O, separated sEVs were visualised by negative-positive contrast and immunogold labelling as described previously (PMID: 37103858).

Reviewer #1: Please include this justification in the revised version.

We included this justification in the revised manuscript.

(9) For Figures 3C and 3-S1, the authors should include the images used for EV quantification. Considering the concern regarding potential contamination introduced by FBS (concern 1), it is advisable for the authors to employ an independent method to identify EVs, thereby confirming the reliability of the data presented in these figures.

In our revised manuscript, we will provide all the images used for EV quantification in Figure 3C. Given that Figures 3C and 3-S1 show MV-IEVs released by HEK293T-PlamGFP cells, the possible interference by FBS-derived non-fluorescent EVs can be excluded.

Reviewer #1: Please provide all the images.

Original LASX files are provided (DOI: 10.6019/S-BIAD1456).

Reviewer #1: The images raising concerns regarding the contamination of EVs in FBS primarily consist of transmission electron microscopy (TEM) images, namely, Figure 1, 2P-S, S5, and Figure 4 P-U, along with the quantification of EV numbers in Figure 3. These concerns persist despite the use of fluorescent-labeled experiments. While fluorescent-labeled MV-IEVs are conclusively identified as originating from the cells, the MV-IEVs observed in Figure 1, 2P-S, S5, and Figure 4 P-U and Figure 3 may derive from both the cells and FBS.

Large EVs (with diameter >800 nm) derived from FBS were not present in our experiments, as discussed above.

(10) Do the amphiectosomes released from other cell types as well as cells in mouse kidneys or liver contain LC3B positive and CD63 positive ILVs?

Based on our confocal microscopic analysis, in addition the HEK293T-PalmGFP cells, HT29 and HepG2 cells also release similar LC3B and CD63 positive MV-IEVs. Preliminary evidence shows MV-IEV secretion by additional cell types.

The response of Reviewer #1: Please show these data in the revised manuscript. Moreover, do cells in mouse kidneys or liver contain LC3B positive and CD63 positive ILVs?

We have added new confocal microscopic images to Fig2-S3 showing amphiectosomes released also by the H9c2 (ATCC) cardiomyoblast cell line. To preserve the ultrastructure of MV-IEVs in complex organs like kidney and liver, fixation with 4% glutaraldehyde with 1% OsO4 appears to be essential. This fixation does not allow for immune detection to assess LC3B and CD63 positive MV-IEVs in the ultrathin sections.

Reviewer #2 (Public Review):

Summary:

The authors had previously identified that a colorectal cancer cell line generates small extracellular vesicles (sEVs) via a mechanism where a larger intracellular compartment containing these sEVs is secreted from the surface of the cell and then tears to release its contents. Previous studies have suggested that intraluminal vesicles (ILVs) inside endosomal multivesicular bodies and amphisomes can be secreted by the fusion of the compartment with the plasma membrane. The 'torn bag mechanism' considered in this manuscript is distinctly different because it involves initial budding off of a plasma membrane-enclosed compartment (called the amphiectosome in this manuscript, or MV-IEV). The authors successfully set out to investigate whether this mechanism is common to many cell types and to determine some of the subcellular processes involved.

The strengths of the study are:

(1) The high-quality imaging approaches used, seem to show good examples of the proposed mechanism.

(2) They screen several cell lines for these structures, also search for similar structures in vivo, and show the tearing process by real-time imaging.

(3) Regarding the intracellular mechanisms of ILV production, the authors also try to demonstrate the different stages of amphiectosome production and differently labelled ILVs using immuno-EM.

Several of these techniques are technically challenging to do well, and so these are critical strengths of the manuscript.

The weaknesses are:

(1) Most of the analysis is undertaken with cell lines. In fact, all of the analysis involving the assessment of specific proteins associated with amphiectosomes and ILVs are performed in vitro, so it is unclear whether these processes are really mirrored in vivo. The images shown in vivo only demonstrate putative amphiectosomes in the circulation, which is perhaps surprising if they normally have a short half-life and would need to pass through an endothelium to reach the vessel lumen unless they were secreted by the endothelial cells themselves.

Our previous results analyzing PFA-fixed, paraffin embedded sections of colorectal cancer patients provided direct evidence that MV-IEV secretion also occurs in humans *in vivo* (PMID: 31007874). Regarding your comment on the presence of amphiectosomes in the circulation despite their short half-lives, we would like to point out that Fig1.X shows a circulating lymphocyte which releases MV-IEV within the vessel lumen. Furthermore, in the revised manuscript, an additional Fig.1-S1 is provided. Here, we show the release of MV-IEVs both by an endothelial and a sub-endothelial cell (Fig.1-S1G). In addition, these images show the simultaneous presence of MV-IEVs and sEVs in the circulation (Fig.1-S1.A,C,D,H and I). The transmission electron micrographs of mouse kidney and liver sections provide additional evidence that the MV-IEVs are released by different types of cells, and the “torn bag release” also takes place in vivo (Fig.1.V).

(2) The analysis of the intracellular formation of compartments involved in the secretion process (Figure 2-S5) relies on immuno-EM, which is generally less convincing than high-/super-resolution fluorescence microscopy because the immuno-labelling is inevitably very sporadic and patchy. High-quality EM is challenging for many labs (and seems to be done very well here), but high-/super-resolution fluorescence microscopy techniques are more commonly employed, and the study already shows that these techniques should be applicable to studying the intracellular trafficking processes.

As you suggested, in the revised manuscript, we present additional super-resolution microscopy (STED) data. The intracellular formation of amphisomes, the fragmentation of LC3B-positive membranes and the formation of LC3B-positive ILVs were captured (Fig. 3B-F).

(3) One aspect of the mechanism, which needs some consideration, is what happens to the amphisome membrane, once it has budded off inside the amphiectosome. In the fluorescence images, it seems to be disrupted, but presumably, this must happen after separation from the cell to avoid the release of ILVs inside the cell. There is an additional part of Figure 1 (Figure 1Y onwards), which does not seem to be discussed in the text (and should be), that alludes to amphiectosomes often having a double membrane.

We agree with your comment regarding the amphisome membrane and we added a sentence to the Discussion of the revised manuscript. Fig1Y onwards is now discussed in the manuscript. In addition, we labelled the surface of living HEK293 cells with wheat germ agglutinin (WGA), which binds to sialic acid and N-acetyl-D-glucosamine. After removing the

unbound WGA by washes, the cells were cultured for an additional 3 hours, and the release of amphiectosomes was studied. The budding amphiectosome had WGA positive membrane providing evidence that the external limiting membrane had a plasma membrane origin (Fig.3G)

(4) The real-time analysis of the amphiectosome tearing mechanism seemed relatively slow to me (over three minutes), and if this has been observed multiple times, it would be helpful to know if this is typical or whether there is considerable variation.

Thank you for this comment. In the revised manuscript, we highlight that the first released LC3 positive ILV was detected as early as within 40 sec.

Overall, I think the authors have been successful in identifying amphiectosomes secreted from multiple cell lines and demonstrating that the ILVs inside them have at least two origins (autophagosome membrane and late endosomal multivesicular body) based on the markers that they carry. The analysis of intracellular compartments producing these structures is rather less convincing and it remains unclear what cells release these structures in vivo.

I think there could be a significant impact on the EV field and consequently on our understanding of cell-cell signalling based on these findings. It will flag the importance of investigating the release of amphiectosomes in other studies, and although the authors do not discuss it, the molecular mechanisms involved in this type of 'ectosomal-style' release will be different from multivesicular compartment fusion to the plasma membrane and should be possible to be manipulated independently. Any experiments that demonstrate this would greatly strengthen the manuscript.

We appreciate these comments of the reviewer. Experiments are on their way to elucidate the mechanism of the “ectosomal style” exosome release and will be the topic of our next publication.

In general, the EV field has struggled to link up analysis of the subcellular biology of sEV secretion and the biochemical/physical analysis of the sEVs themselves, so from that perspective, the manuscript provides a novel angle on this problem.

Reviewer #3 (Public Review):

Summary:

In this manuscript, the authors describe a novel mode of release of small extracellular vesicles. These small EVs are released via the rupture of the membrane of so-called amphiectosomes that resemble "morphologically" Multivesicular Bodies.

These structures have been initially described by the authors as released by colorectal cancer cells (<https://doi.org/10.1080/20013078.2019.1596668>). In this manuscript, they provide experiments that allow us to generalize this process to other cells. In brief, amphiectosomes are likely released by ectocytosis of amphisomes that are formed by the fusion of multivesicular endosomes with autophagosomes. The authors propose that their model puts forward the hypothesis that LC3 positive vesicles are formed by "curling" of the autophagosomal membrane which then gives rise to an organelle where both CD63 and LC3 positive small EVs co-exist and would be released then by a budding mechanism at the cell surface that appears similar to the budding of microvesicles /ectosomes. Very correctly the authors make the distinction from migrasomes because these structures appear very similar in morphology.

Strengths:

The findings are interesting despite that it is unclear what would be the functional relevance of such a process and even how it could be induced. It points to a novel mode of release of extracellular vesicles.

Weaknesses:

This reviewer has comments and concerns concerning the interpretation of the data and the proposed model. In addition, in my opinion, some of the results in particular micrographs and immunoblots (even shown as supplementary data) are not of quality to support the conclusions.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Highlight MV-IEV, ILV and limiting membrane in Figure-1G, N, and U.

Based on the suggestion, we revised Figure1

(2) Figure 1-Y-AF are not mentioned in the text.

In the revised manuscript, we discuss Figure 1Y-AF

(3) The term "IEVs" in Figure 2-S2 is not defined.

We modified the figure legend: we changed MV-IEV to amphiectosome

(4) Need to quantify co-localization in Figure 2-S2.

As suggested, we carried out the co-localisation analysis (Fig2-S2I), and Fig2-S2 was re-edited

Reviewer #2 (Recommendations For The Authors):

I have two recommendations for improving the manuscript through additional experiments:

(1) I think the description of the intracellular processes taking place in order to form amphiectosomes would be much stronger if some super-resolution imaging could be undertaken. This should label the different compartments before and after fusion with specific markers that highlight the protein signature of the different limiting and ILV membranes much more clearly than immuno-EM. It will also help in characterising the double-membrane structure of amphiectosomes at the point of budding and reveal whether the patchy labelling of the inner membrane emerges after amphiectosome release (the schematic model currently suggests that it happens before).

Thank you for your suggestion. STED microscopy was applied and results are shown in new Fig3 and the schematic model was modified accordingly.

(2) The implications of the manuscript would be more wide-ranging if the authors could test genetic manipulations that are believed to block exosome or ectosome release, eg. Rab27a or Arrdc1 knockdown. This may allow them to determine whether MV-IEVs can be released independently of the classical exosome release mechanism because they use a different route to be released from the plasma membrane. This experiment is not essential, but I think it would start to address the core regulatory mechanisms involved, and if successful, would easily allow the authors to determine the ratio of CD63-positive sEVs being secreted via classical versus amphiectosome routes.

The suggestion is very valuable for us and these studies are being performed in a separate project.

I think there are several other ways in which the manuscript could be improved to better explain some of the approaches, findings and interpretation:

(1) Include some explanation in the text of certain key tools, particularly:

a. Palm-GFP and whether its expression might alter the properties of the plasma membrane since this is used in a lot of experiments and is the only marker that seems to uniformly label the outer membrane of amphiectosomes. One concern might be that its expression drives amphiectosome secretion.

We found evidence for amphiectosome release also in the case of several different cells not expressing Palm-GFP. We believe, this excludes the possibility that Palm-GFP expression is the inducer of the amphiectosome release. Both by fluorescent and electron microscopy, the Palm-GFP non expressing cells showed very similar MV-IEVs. In addition, in the case of non-transduced HEK293 and fluorescent WGA-binding, we made similar observations.

b. Lactadherin - does this label the amphiectosomes after their release or does the wash-off step mean that it only labels cells, which subsequently release amphiectosomes?

Lactadherin labels the amphiectosomes after their release and fixation. Living cells cannot be labelled by lactadherin as PS is absent in the external plasma membrane layer of living cells. We used WGA on HEK293 cells to further support the plasma membrane origin of the external membrane of amphiectosomes.

(2) Explain the EM and confocal imaging approaches more clearly. Most importantly, is a 3D reconstruction always involved to confirm that 'separated' amphiectosomes are not joined to cells in another Z-plane.

Thank you for your suggestion. We have modified the manuscript accordingly

(3) Presenting triple-labelled images with red, green and yellow channels does not allow individual labelling to be determined without single-channel images and even then, it is much more informative to use three distinguishable colours that make a different colour with overlap, eg. CMY? Fig.2_S2D and E do not display individual channels, so definitely need to be changed.

In case of Fig.2_S2D, we now show the individual channels, the earlier E image has been removed. In case of the STED images, CMY colors had been used, as you suggested.

(4) Please discuss in the text the data in Figure 1Y onwards concerning single/double membranes on MV-IEVs.

In the revised manuscript, we discuss the question on single/double membranes and we refer to Figure 1Y-AF

(5) On line 162, reword 'intraluminal TSPAN4 only' to 'one in which TSPAN4 is only intraluminal' to make it clear that other proteins are also marking the intraluminal region, not TSPAN4 only.

We modified the text accordingly.

(6) Points for further discussion and further conclusions:

a. In vivo experiments - discuss the limitations of this part of the analysis - it seems that none of the amphiectosome markers have been analysed in this part of the study and the MV-IEVs are only in the circulation.

b. Can the authors give any further indication of the levels of MV-IEVs relative to free sEVs from any of their studies?

Using our current approach, it is not possible to determine the levels of MV-IEVs to free sEV. Without analyzing serial ultrathin sections, determination of the relative ratio of MV-IEVs and sEVs would depend on the actual section plane. In future projects, we will determine the ratio of LC3 positive and negative sEVs by single EV analysis techniques (such as SP-IRIS). In the revised manuscript, additional TEM images are included to provide evidence for the simultaneous presence of sEVs and MV-IEVs and MV-IEVs both inside and outside of the circulation.

c. Please discuss the single versus double membrane issue (relating to experiments proposed above).

We discuss this question in more details in the revised manuscript.

d. Please point out that the release mechanism (plasma membrane budding) will involve different molecular mechanisms to establish exosome release, and this might provide a route to determine relative importance.

We are currently running a systemic analysis of the release mechanism of amphiectosomes, and this will be the topic of a separate manuscript.

Reviewer #3 (Recommendations For The Authors):

** The model is not supported.*

** The data is not of quality.*

** The appropriate methods are not exploited.*

We are sorry, we cannot respond to these unsupported critiques.

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